

ECO 329 ECONOMIC STATISTICS (36110)

Lecture Notes

Fall 2025

5 Sampling

Definition 5.1. A **(sample) statistic** is a quantity whose value can be calculated from sample data.

- A statistic is a r.v. because it is a function of the sample observations - and there is uncertainty associated with sample observations prior to the obtainment of the data.
- A statistic is usually denoted with uppercase letters, and the value calculated from some sample is usually denoted with lowercase letters.
 - For instance, sample mean is $\bar{X}$, and a concrete value of it calculated from some sample is $\bar{x}$.
- The probability distribution of a sample statistic is called its **sampling distribution**, which is frequently used in inferential statistics.

Definition 5.2. Random variables $X_1, X_2, \dots, X_n$ are said to form a **random sample** of size $n \in \mathbb{N}_+$ if X_i 's

1. are mutually independent and
 2. are with the same distribution, i.e., identically distributed.
- Throughout the rest of this course, unless specified otherwise, for $\{X_i\}_{i=1}^n$, a random sample of size $n \in \mathbb{N}_+$, denote its sample mean as $\bar{X} = \sum_{i=1}^n X_i$ and its sample variance as $S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$.

Proposition 5.3. For a random sample $\{X_i\}_{i=1}^n$ from a distribution with mean μ and variance σ^2 , then

$$E(\bar{X}) = \mu \text{ and } V(\bar{X}) = \frac{\sigma^2}{n}.$$

- Proof idea: Direct calculation by definitions.
- The standard deviation of a sample statistic is usually called the **standard error** of the statistic. For example, the standard error of the sample mean above is $\sqrt{V(\bar{X})} = \frac{\sigma}{\sqrt{n}}$.

Proposition 5.4. For a random sample $\{X_i\}_{i=1}^n$ from $N(\mu, \sigma^2)$, then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

- Proof idea: By Proposition 4.3.4.5 and Proposition 5.3.

Theorem 5.5. [Central Limit Theorem (CLT)] For a random sample $\{X_i\}_{i=1}^n$ from a distribution with mean μ and (finite) variance σ^2 , when n is sufficiently large, $\bar{X}$ approximately follows $N\left(\mu, \frac{\sigma^2}{n}\right)$.

- Or formally, as $n \rightarrow \infty$, $\sqrt{n}(\bar{X} - \mu)$ converges (in distribution) to $N(0, \sigma^2)$.
 - Or equivalently, $\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$ in the limit.
- Proof: not required.
- A rule of thumb in terms of using such a CLT is $n > 30$.
- When σ is unknown, CLT also gives that $\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \sim N(0, 1)$ in the limit.

Proposition 5.6. For a random sample $\{X_i\}_{i=1}^n$ from $N(\mu, \sigma^2)$,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2 \text{ and } \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \sim t_{n-1}.$$

- Proof: not required.

6 Estimation

6.1 Point Estimation

Definition 6.1.1. A **point estimator** of a (population) parameter θ is a sample statistic designed to yield a single value - i.e., a point - that estimates θ when the statistic is evaluated using some specific sample data.

- The value (which is a single number) of a point estimator yielded from some specific sample data is called a **point estimate**.

Definition 6.1.2. A point estimator $\hat{\theta}$ is said to be an **unbiased estimator** of θ if $E(\hat{\theta}) = \theta$ for every possible value of θ .

- Correspondingly, $\hat{\theta}$ is **biased** if $E(\hat{\theta}) \neq \theta$.

Definition 6.1.3. The **bias** of $\hat{\theta}$ is $Bias(\hat{\theta}) = E(\hat{\theta}) - \theta$.

- By contrast, $\hat{\theta} - \theta$ is called the **sampling error** of $\hat{\theta}$.

- The population parameter θ is a constant - not a random variable - implying that $E(\theta) = \theta$.

Proposition 6.1.4. For a random sample $\{X_i\}_{i=1}^n$ from a distribution with mean μ , $\bar{X}$ is an unbiased estimator of μ , i.e.,

$$E(\bar{X}) = \mu.$$

- Proof: Direct calculation by definitions.
- Specially, if the random sample is from $b(m, p)$, $E(\bar{X}) = mp$, implying $E(\frac{\bar{X}}{m}) = p$.

Proposition 6.1.5. For a random sample $\{X_i\}_{i=1}^n$ from a distribution with mean μ and variance σ^2 , $(\hat{\sigma}^2 =) S^2$ is a unbiased estimator of σ^2 , i.e.,

$$E(S^2) = \sigma^2.$$

- Proof: Direct calculation by definitions.

Definition 6.1.6. The **standard error** of an estimator $\hat{\theta}$ is its standard deviation $\sigma_{\hat{\theta}} = \sqrt{V(\hat{\theta})}$.

- $\sigma_{\hat{\theta}}$ evaluates the magnitude of a typical/representative deviation between an estimate and the true value of the population parameter - i.e., in general/on average, how dispersed is the difference between the value of an estimate and that of population parameter.
- If $\sigma_{\hat{\theta}}$ itself involves some unknown values (e.g., population parameters) that can be estimated, one may replace the values with the relevant estimates to get the **estimated standard error** $\hat{\sigma}_{\hat{\theta}}$.

Definition 6.1.7. The **mean squared error** of an estimator $\hat{\theta}$ of θ is

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2].$$

- By definitions, $E[(\hat{\theta} - \theta)^2] = [Bias(\hat{\theta})]^2 + V(\hat{\theta})$.
 - Accordingly, there is a trade-off between bias and variance in terms of selection of estimators - and, at least in some extreme situations, a biased estimator may be more preferable than an unbiased one.

Definition 6.1.8. For two unbiased estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ of θ , if $V(\hat{\theta}_1) < V(\hat{\theta}_2)$, then $\hat{\theta}_1$ is **relatively more efficient** than $\hat{\theta}_2$.

- Efficiency can only be used for comparing unbiased estimators. By contrast, MSE can be used in the comparison among estimators that are possibly biased.

Definition 6.1.9. An estimator of θ is said to be a **linear estimator** of it if the estimator could be written as a linear combination of sample observations.

Definition 6.1.10. Among all linear unbiased estimators of θ , one with the minimum variance is called a **best linear unbiased estimator (BLUE)** of θ .

Proposition 6.1.11. For a random sample $\{X_i\}_{i=1}^n$ from a population with mean μ and variance σ^2 , $\bar{X}$ is a BLUE of μ .

- Proof idea (not required): Cauchy–Schwarz inequality or Jensen’s inequality.

6.2 Interval Estimation

Definition 6.2.1. An **interval estimator** of a (population) parameter θ is a random interval of the form

$$P(\theta \in [L, U]) = 1 - \alpha,$$

where L and U are functions of sample (observations) and possibly other known parameter values.

- In other words, L and U can be treated as r.v.’s whose values depend on the drawn sample.
- The interval estimator investigated in this course is called **confidence interval (CI)**, which can be expressed in any of the three following equivalent ways:
 - $[L, U]$ is a $1 - \alpha$ confidence interval for θ ;
 - $\theta \in [L, U]$ with $1 - \alpha$ confidence (level);
 - $C(\theta \in [L, U]) = 1 - \alpha$.
- The **confidence level** $1 - \alpha \in [0, 1]$, which appears as a probability in the definition above, shouldn’t be interpreted as “the probability that the true parameter value lies in a calculated CI is $1 - \alpha$ ”.
 - Instead, it is the fraction of the CI’s yielded from all possible samples that contain the true parameter value.

Definition 6.2.2. A $1 - \alpha$ confidence interval $[L, U]$ for θ is a **one-sided confidence interval** if $L \rightarrow -\infty$ or $U \rightarrow +\infty$.

- A one-sided confidence interval $1 - \alpha$ puts a probability of α on one tail and 0 on the other - so that the CI takes the shape of $(-\infty, U]$ (called a lower-sided CI) or $[L, +\infty)$ (called an upper-sided CI).
- By contrast, a CI $[L, U]$ is two-sided when L and U are both finite.

Method 6.2.3. Confidence interval for mean of a normal population $N(\mu, \sigma^2)$ with σ^2 known.

1. With a random sample $\{X_i\}_{i=1}^n$ drawn from the population, by Proposition 4.3.4.5 and Proposition 4.2.2.6, we have $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$, yielding

$$\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1).$$

2. Thus, to construct a $1 - \alpha$ CI for μ , we have the following fact to start with:

$$P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \in [z_{1-\alpha+\beta}, z_\beta]\right) = 1 - \alpha \text{ for all } \beta \in [0, \alpha], \text{ i.e.,}$$

$$P\left(\mu \in \left[\bar{X} - z_\beta \frac{\sigma}{\sqrt{n}}, \bar{X} - z_{1-\alpha+\beta} \frac{\sigma}{\sqrt{n}}\right]\right) = 1 - \alpha \text{ for all } \beta \in [0, \alpha].$$

3. Accordingly, we can get

- (a) a $1 - \alpha$ lower-sided CI for μ by letting $\beta \rightarrow 0$, so that the CI is $(-\infty, \bar{X} + z_\alpha \frac{\sigma}{\sqrt{n}}]$,
 - (b) a $1 - \alpha$ upper-sided CI for μ by letting $\beta \rightarrow \alpha$, so that the CI is $[\bar{X} - z_\alpha \frac{\sigma}{\sqrt{n}}, +\infty)$,
and
 - (c) a symmetric two-sided CI for μ by letting $\beta = \frac{\alpha}{2}$, so that the CI is $[\bar{X} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}]$.
- The symmetry of normal distributions is applied in simplifying the expressions of the CI's so that z_α and $z_{\frac{\alpha}{2}}$ are the only percentages used in the expressions.
 - For any of the confidence intervals above, its length is decreasing in the sample size n . Thus, if we want a “more precise” - in the sense of being shorter - CI, one effective way is to increase the sample size.
 - When the sampling distribution used to construct CI is symmetric and unimodal (in the case above, normal distributions), the two-sided $1 - \alpha$ CI that (is symmetric in the sense that it) puts $\frac{\alpha}{2}$ on each of the two tails is the shortest among all two-sided $1 - \alpha$ CI's.
 - Throughout this course, unless specified otherwise, the discussion regarding two-sided $1 - \alpha$ CI's focuses on those with $\frac{\alpha}{2}$ put on each of the two tails.

Method 6.2.4. Confidence interval for mean of a population with mean μ , unknown variance σ^2 , and large sample size.

1. With a random sample $\{X_i\}_{i=1}^n$ drawn from the population, because the sample size n is large - which, technically and approximately, means that $n \rightarrow +\infty$ - because S^2 consistently estimates σ^2 , combined with a CLT, we have that approximately

$$\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \sim N(0, 1).$$

2. Thus, to construct a $1 - \alpha$ CI for μ , we have the following fact to start with:

$$P\left(\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \in [z_{1-\alpha+\beta}, z_\beta]\right) = 1 - \alpha \text{ for all } \beta \in [0, \alpha], \text{ i.e.,}$$

$$P\left(\mu \in \left[\bar{X} - z_\beta \frac{S}{\sqrt{n}}, \bar{X} - z_{1-\alpha+\beta} \frac{S}{\sqrt{n}}\right]\right) = 1 - \alpha \text{ for all } \beta \in [0, \alpha].$$

3. Accordingly, we can get

- (a) a $1 - \alpha$ lower-sided CI for μ by letting $\beta \rightarrow 0$, so that the CI is $(-\infty, \bar{X} + z_\alpha \frac{S}{\sqrt{n}}]$,
 - (b) a $1 - \alpha$ upper-sided CI for μ by letting $\beta \rightarrow \alpha$, so that the CI is $[\bar{X} - z_\alpha \frac{S}{\sqrt{n}}, +\infty)$,
and
 - (c) a symmetric two-sided CI for μ by letting $\beta = \frac{\alpha}{2}$, so that the CI is $[\bar{X} - z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}}, \bar{X} + z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}}]$.
- With large sample, the population need not be normal for one to apply the method above.
 - For example, for a Bernoulli population with success probability p , one can construct a CI with

$$\frac{\bar{X} - p}{\sqrt{\frac{\bar{X}(1-\bar{X})}{n}}} \sim N(0, 1).$$

- * Here, a rule of thumb of applying such a CLT to a Bernoulli population is $np \geq 10$ and $n(1-p) \geq 10$.
- If the population variance σ^2 is known in a large-sample situation, one can still apply the above method with each S directly replaced with σ .

Method 6.2.5. Confidence interval for mean of a normal population $N(\mu, \sigma^2)$ with σ^2 unknown and small sample size.

1. With a random sample $\{X_i\}_{i=1}^n$ drawn from the normal population, by Proposition 5.6,

$$\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \sim t_{n-1}.$$

2. Thus, to construct a $1 - \alpha$ CI for μ , we have the following fact to start with:

$$P\left(\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \in [t_{1-\alpha+\beta, n-1}, t_{\beta, n-1}]\right) = 1 - \alpha \text{ for all } \beta \in [0, \alpha], \text{ i.e.,}$$

$$P\left(\mu \in \left[\bar{X} - t_{\beta, n-1} \frac{S}{\sqrt{n}}, \bar{X} - t_{1-\alpha+\beta, n-1} \frac{S}{\sqrt{n}}\right]\right) = 1 - \alpha \text{ for all } \beta \in [0, \alpha].$$

3. Accordingly, we can get

- (a) a $1 - \alpha$ lower-sided CI for μ by letting $\beta \rightarrow 0$, so that the CI is $(-\infty, \bar{X} + t_{\alpha, n-1} \frac{S}{\sqrt{n}}]$,
- (b) a $1 - \alpha$ upper-sided CI for μ by letting $\beta \rightarrow \alpha$, so that the CI is $[\bar{X} - t_{\alpha, n-1} \frac{S}{\sqrt{n}}, +\infty)$,
and
- (c) a symmetric two-sided CI for μ by letting $\beta = \frac{\alpha}{2}$, so that the CI is $[\bar{X} - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}, \bar{X} + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}]$.

Method 6.2.6. Confidence interval for variance of a normal population $N(\mu, \sigma^2)$.

1. With a random sample $\{X_i\}_{i=1}^n$ drawn from the normal population, by Proposition 5.6,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

2. Thus, to construct a $1 - \alpha$ CI for μ , we have the following fact to start with:

$$P\left\{\frac{(n-1)S^2}{\sigma^2} \in [\chi_{1-\alpha+\beta, n-1}^2, \chi_{\beta, n-1}^2]\right\} = 1 - \alpha \text{ for all } \beta \in [0, \alpha], \text{ i.e.,}$$

$$P\left\{\sigma^2 \in \left[\frac{(n-1)S^2}{\chi_{\beta, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\alpha+\beta, n-1}^2}\right]\right\} = 1 - \alpha \text{ for all } \beta \in [0, \alpha].$$

3. Accordingly, we can get

- (a) a $1 - \alpha$ “lower-sided” CI for μ by letting $\beta \rightarrow 0$, so that the CI is $(0, \frac{(n-1)S^2}{\chi_{1-\alpha, n-1}^2}]$,
- (b) a $1 - \alpha$ upper-sided CI for μ by letting $\beta \rightarrow \alpha$, so that the CI is $[\frac{(n-1)S^2}{\chi_{\alpha, n-1}^2}, +\infty)$, and
- (c) a symmetric two-sided CI for μ by letting $\beta = \frac{\alpha}{2}$, so that the CI is $[\frac{(n-1)S^2}{\chi_{\frac{\alpha}{2}, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2}]$.
 - Because the χ^2 distribution is not symmetric, the above two-sided CI is not optimal in that it is not the shortest among all CI's with the same confidence level.
 - However, we usually work with such a symmetric CI for the following two reasons.
 - i. The value of β that yields a shortest two-sided CI varies by the value of α .
 - ii. Such symmetry makes the CI easily applicable to hypothesis testing (more to say later).

Example 5.A: Normal Population with Known Variance

Consider a sports equipment manufacturer whose 2-lb dumbbells are with weights normally distributed with mean μ and variance 0.0016 (unit: lb). A random sample of 2-lb dumbbells $\{X_i\}_{i=1}^{25}$ with sample size 25 is obtained from the manufacturer, where X_i 's are the weights of the dumbbells.

1. Calculate the expectation of the sample mean $\bar{X}$.
2. Calculate the standard error of $\bar{X}$.
3. Write down the sampling distribution of $\bar{X}$.

Suppose throughout of the rest of this problem that $\bar{X} = 2.003$.

4. Construct a symmetric 95% two-sided confidence interval for the population mean μ .
5. Construct a 95% lower-sided confidence interval for the population mean μ .
6. Construct a 90% upper-sided confidence interval for the population mean μ .

Note: We will need the following z critical values in this problem: $z_{0.1} = 1.2816$, $z_{0.05} = 1.6449$, $z_{0.025} = 1.96$.

Example 5.B: Large Sample, Unknown Population

Consider an assembly line of a chip manufacturer whose hourly output is with mean μ and variance σ^2 , both of which are unknown. A random sample of hourly outputs $\{X_i\}_{i=1}^{100}$ with sample size 100 is obtained from the assembly line.

1. Write down the limiting distribution regarding the sample mean $\bar{X}$ yielded from central limit theorem.

Suppose throughout the rest of this problem that $\bar{X} = 1500$ and sample variance S^2 equals 400.

2. Construct a symmetric 95% two-sided confidence interval for the population mean μ .
3. Construct a 95% lower-sided confidence interval for the population mean μ .
4. Construct a 90% upper-sided confidence interval for the population mean μ .

Note: We will need the following z critical values in this problem: $z_{0.1} = 1.2816$, $z_{0.05} = 1.6449$, $z_{0.025} = 1.96$.

Example 5.C: Small Sample, Normal Population with Unknown Variance

Consider a pizza restaurant whose 10-inch pizzas are with diameters normally distributed with mean μ and variance σ^2 (unit: inch), both of which are unknown. A random sample of 10-inch pizzas $\{X_i\}_{i=1}^{16}$ with sample size 16 is obtained from the restaurant, where X_i 's are the diameters of the pizzas. The sample mean $\bar{X}$ equals 10.2 and sample variance S^2 equals 0.04.

1. Write down the statistic and its sampling distribution that can be used for constructing confidence intervals for μ .
2. Construct a symmetric 95% two-sided confidence interval for μ .
3. Construct a 95% lower-sided confidence interval for μ .
4. Construct a 90% upper-sided confidence interval for μ .

Note: We will need the following t critical values in this problem: $t_{0.1,15} = 1.3406$, $t_{0.05,15} = 1.7531$, $t_{0.025,15} = 2.1314$.

5. Write down the statistic and its sampling distribution that can be used for constructing confidence intervals for σ^2 .
6. Construct a symmetric 95% two-sided confidence interval for σ^2 .
7. Construct a 95% lower-sided confidence interval for σ^2 .
8. Construct a 90% upper-sided confidence interval for σ^2 .

Note: We will need the following χ^2 critical values in this problem: $\chi_{0.05,15}^2 = 24.9958$, $\chi_{0.1,15}^2 = 22.3071$, $\chi_{0.9,15}^2 = 8.5468$, $\chi_{0.95,15}^2 = 7.2609$.

Example 6.A: Point Estimation with a Uniform Distribution

Consider a bus route where buses depart with a delay of $X \sim [0, 2a]$ minutes ($a > 0$). A random sample of departure delays $\{X_i\}_{i=1}^{20}$ with sample size 20 is obtained from the bus route.

1. Prove that $\bar{X}$ is an unbiased estimator of a .
2. Prove that S^2 is an unbiased estimator of $\frac{a^2}{3}$.
3. Calculate $MSE(\bar{X})$.

Note: $MSE(\bar{X})$ would be a function of the parameter a .

4. Calculate the estimated standard error of $\bar{X}$.

Example 6.B: Large sample, Bernoulli population

Consider a university where the students are either for or against a certain university-wide regulation. The fraction of students who are for the regulation is p . A random sample of students $\{X_i\}_{i=1}^{400}$ with sample size 400 is obtained from the students in the university, where for each student with label $i \in \{1, 2, \dots, 400\}$, $X_i = 1$ if the student is for the regulation and $x_i = 0$ if the student is against the regulation. The sample mean $\bar{X}$ equals 0.8.

1. Write down the limiting distribution regarding the sample mean $\bar{X}$ yielded from central limit theorem that can be used for constructing confidence intervals for p .
2. Construct a symmetric 95% two-sided confidence interval for p .
3. Construct a 95% lower-sided confidence interval for p .
4. Construct a 90% upper-sided confidence interval for p .

Note: We will need the following z critical values in this problem: $z_{0.1} = 1.2816$, $z_{0.05} = 1.6449$, $z_{0.025} = 1.96$.